

Tutorial 4. Shading and Photometric Stereo

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Interactive lab: cv-tutorial-04-lab.chenrui-fan.chatgpt.site

Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Scope. This tutorial practises Lecture 4 on shading, shape from shading, integrability and photometric stereo: relating normals to depth, reading the reflectance map, recovering albedo and normals from several lights, and formulating shape from shading. Q7 applies the Lambertian model to a nearby point light; its part (b) goes beyond the slides. Slide numbers refer to the Lecture 4 slides lec04a. The solution sheet repeats each question before its answer.

Conventions. We use the frame of slides 9-12: x and y span the image plane, and $+z$ points toward the camera. The visible surface is $(x, y, z(x, y))$, seen by an orthographic camera. A light direction $\mathbf{s}$ is the unit vector from the surface point toward the light source. In this tutorial every light is on the camera side of the surface, so $s_3 > 0$. As on slide 9, the normal is written as

$$\mathbf{N} = (p, q, 1)^\top, \quad p = -\frac{\partial z}{\partial x}, \quad q = -\frac{\partial z}{\partial y}, \quad \mathbf{n} = \frac{\mathbf{N}}{\|\mathbf{N}\|_2}.$$

Slides 25 and 31-33 draw the z -axis the other way, pointing away from the camera, so the same normal has third component -1 there. Both drawings are consistent: in each of them, $\nabla z = -(n_1/n_3, n_2/n_3)^\top$. Some texts define $p = \partial z / \partial x$ instead, which reverses the signs of p and q ; check the definition before comparing formulas. If image rows increase downward, convert row indices to y first.

Key concepts.

- *Lambertian shading* (slides 7-8). A distant light with unit direction $\mathbf{s}$ and intensity L gives a point with albedo ρ and unit normal $\mathbf{n}$ the brightness $I = \rho L \max(0, \langle \mathbf{n}, \mathbf{s} \rangle)$. For a point that faces the light, this is the slide formula $I = \rho \langle \mathbf{n}, \mathbf{s} \rangle L$; a point with $\langle \mathbf{n}, \mathbf{s} \rangle \leq 0$ is in attached shadow and has $I = 0$. The viewing direction does not appear.
- *Gradient space* (slide 10). Each orientation that faces the camera is one point (p, q) of the plane $z = 1$.
- *Reflectance map* (slides 10-12). With $\rho = L = 1$, as on slide 8, the brightness of an orientation (p, q) is $R(p, q) = \langle \mathbf{n}, \mathbf{s} \rangle = \cos \theta_i$ if it faces the light, and 0 otherwise. R is largest, $R = 1$, when $\mathbf{n} = \mathbf{s}$ (slide 12). An iso-brightness contour collects all orientations with the same brightness.
- *Integrability*. A field (p, q) is the slope field of a smooth depth map only if $p_y = q_x$.
- *Photometric stereo* (slides 14-20). Camera and object stay fixed while known lights change; each pixel gives a linear system for $\mathbf{g} = \rho \mathbf{n}$.
- *Normal integration* (slides 25-28). Least squares turns a normal map into a depth map, up to an additive constant.
- *Shape from shading* (slides 30-34). One image plus prior constraints, solved by minimising an energy.

Surface normals and shading

Focus: normals and depth slopes, integrability, the reflectance map and the single-image ambiguity.

Q1. Normals, depth and integrability (slides 9, 25-28).

- (a) Use the tangent vectors of $(x, y, z(x, y))$ to show that $\mathbf{N} = (p, q, 1)^\top$ with $p = -z_x$ and $q = -z_y$ is normal to the surface. Express ∇z through a measured unit normal $\mathbf{n}$.
- (b) Compute p , q and $\mathbf{n}$ for the plane $z = 2x - y + 3$. Why is n_1 negative?
- (c) Show that a field (p, q) can come from a smooth depth map only if $p_y = q_x$. Test $(p, q) = (y, x)$ and $(p, q) = (y, 0)$.
- (d) On a pixel grid, write the least-squares problem that recovers z from a normal map with finite differences. Why must one depth value, or the mean depth, be fixed? What goes wrong where $n_3 \rightarrow 0$?

Q2. Reflectance map and the single-image ambiguity (slides 7-8, 10-13). A Lambertian surface with $\rho = 1$ is lit by a distant light of intensity $L = 1$. The light direction is $\mathbf{s} \propto (p_s, q_s, 1)^\top = (1, 0, 1)^\top$.

- (a) Which quantities change the brightness of a Lambertian point? Why is the viewing direction not one of them?
- (b) Write the reflectance map $R(p, q)$. Where is it maximal, and on which line is it zero?
- (c) A pixel has brightness $I = 1/\sqrt{2}$. Show that every orientation on the parabola $q^2 = 2p$ explains this value, and give two clearly different normals on it. What does this mean for recovering shape from one image?

Photometric stereo

Focus: linear recovery of albedo and normals from known lights; rank, least squares and shadows.

Q3. Minimum number of images (slides 14-16). At one surface point, assume a Lambertian material, a fixed camera and object, and distant lights with known directions $\mathbf{s}_k$ and intensities L_k .

- (a) What is the minimum number of images needed to recover the albedo ρ and the unit normal $\mathbf{n}$? Which condition must the lights satisfy?
- (b) Can the three lights $\mathbf{s}_1 = (0, 0, 1)^\top$, $\mathbf{s}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$ and $\mathbf{s}_3 = \frac{1}{\sqrt{2}}(-1, 0, 1)^\top$ be used? Explain what cannot be recovered.
- (c) Which further assumptions must hold for the linear equations to be valid?

Q4. Solving for albedo and normal (slides 15-16). Use the lights $\mathbf{s}_1 = (0, 0, 1)^\top$, $\mathbf{s}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$ and $\mathbf{s}_3 = \frac{1}{\sqrt{2}}(0, 1, 1)^\top$ with $L_k = 1$. At one pixel the measured linear intensities are

$$\mathbf{I} = \left(\frac{1}{5}, \frac{3\sqrt{2}}{10}, \frac{3\sqrt{2}}{10} \right)^\top.$$

Form and solve the linear system for $\mathbf{g} = \rho\mathbf{n}$. Recover ρ and $\mathbf{n}$, then convert $\mathbf{n}$ to (p, q) and to the depth slopes (z_x, z_y) . Why is $\mathbf{g}$ not a unit vector?

Q5. More than three images, shadows and colour (slides 17-20).

- (a) For $m > 3$ lights, write the least-squares estimate of $\mathbf{g}$. When is it unique, and why do extra images help?
- (b) Slide 20 multiplies the k -th equation by its own intensity I_k . Write the resulting system. What happens to a measurement in shadow, and which bad measurements does this step not remove?
- (c) How can colour images be handled (slide 18)?

Shape from shading

Focus: one image, regularisation, boundary conditions and iterative minimisation.

Q6. Shape from shading as energy minimisation (slides 30-34). Only one image $I(x, y)$ is available, with $\rho = L = 1$ and a known light direction $\mathbf{s}$.

- (a) Write an energy for the unknown field (p, q) with a data term based on $R(p, q)$, a smoothness term and the integrability penalty from Q1. What does each term do? What happens if the smoothness weight is very small or very large?
- (b) Why do slides 31-33 replace (p, q) by (f, g) ? What is known about $\mathbf{n}$ on the occluding boundary (slide 32), and why does it help?
- (c) Slide 33 minimises $E[f, g] = \iint (I - R(f, g))^2 + \lambda(\|\nabla f\|^2 + \|\nabla g\|^2) dx dy$. Derive the gradient equation for f . Then use the approximation of slide 34, $-\Delta f_{i,j} \approx 2(f_{i,j}^{t+1} - \bar{f}_{i,j}^t)$, where $\bar{f}_{i,j}^t$ is the average of the four neighbours, to obtain an iterative update.

Beyond a distant light

Focus: a nearby point light changes the light direction across a flat surface.

Q7. A nearby point light (slides 7-8, 12). The plane $z = 0$ is Lambertian with $\rho = 1$ and is imaged by an orthographic camera, so pixel (x, y) sees the point $\mathbf{X} = (x, y, 0)^\top$. A point light with $L = 1$ is at $\mathbf{A} = (5, 5, 10)^\top$. Because the light is close, use a different unit direction at every point, $\mathbf{s}(x, y) = (\mathbf{A} - \mathbf{X})/\|\mathbf{A} - \mathbf{X}\|_2$, and keep L constant.

- (a) Derive the brightness $B(x, y)$ and find the brightest point. Relate the answer to slide 12.
- (b) *Extension.* The light of a real point source also falls off as $1/\|\mathbf{A} - \mathbf{X}\|_2^2$. Does this change the brightest point? Compare $B(0, 0)/B(5, 5)$ with and without this factor.
- (c) What would the distant-light model predict for this plane?